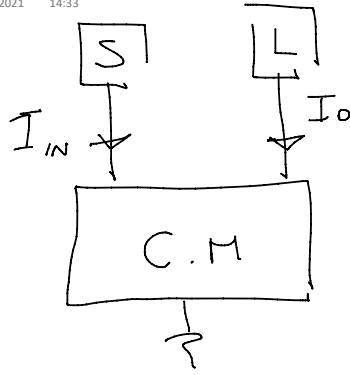


Lecture-1b

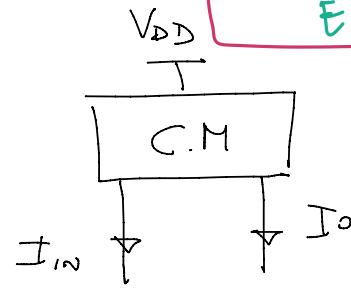
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current mirrors and current references

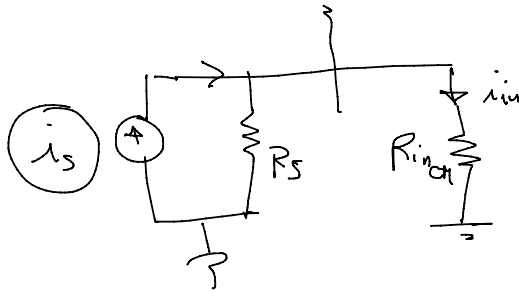
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$$I_O = I_{IN}$$



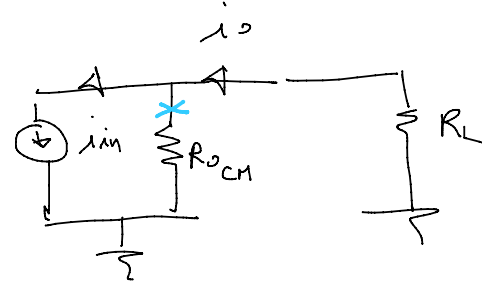
D.S.C



$$R_S \gg R_{inCM}$$

$$i_{in} = \frac{R_S}{R_S + R_{inCM}} i_s$$

Small!

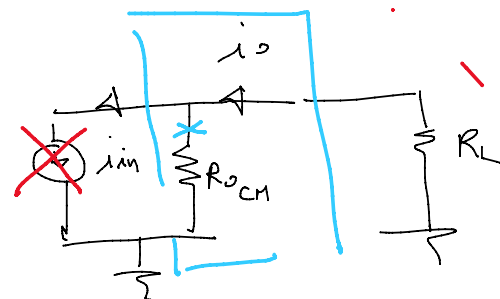
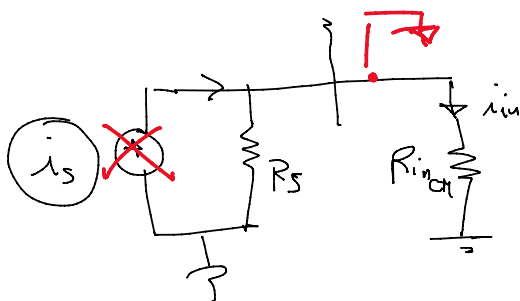


$$R_{oCM} \gg R_L$$

$$i_o = \frac{R_{oCM}}{R_{oCM} + R_L} i_{in}$$

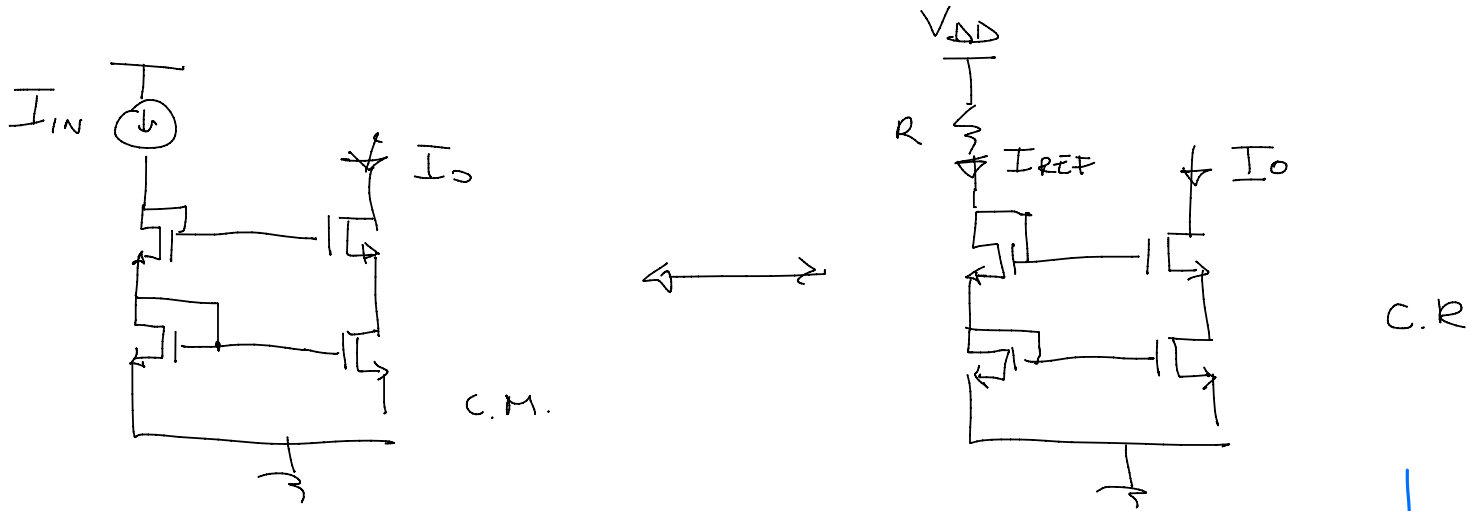
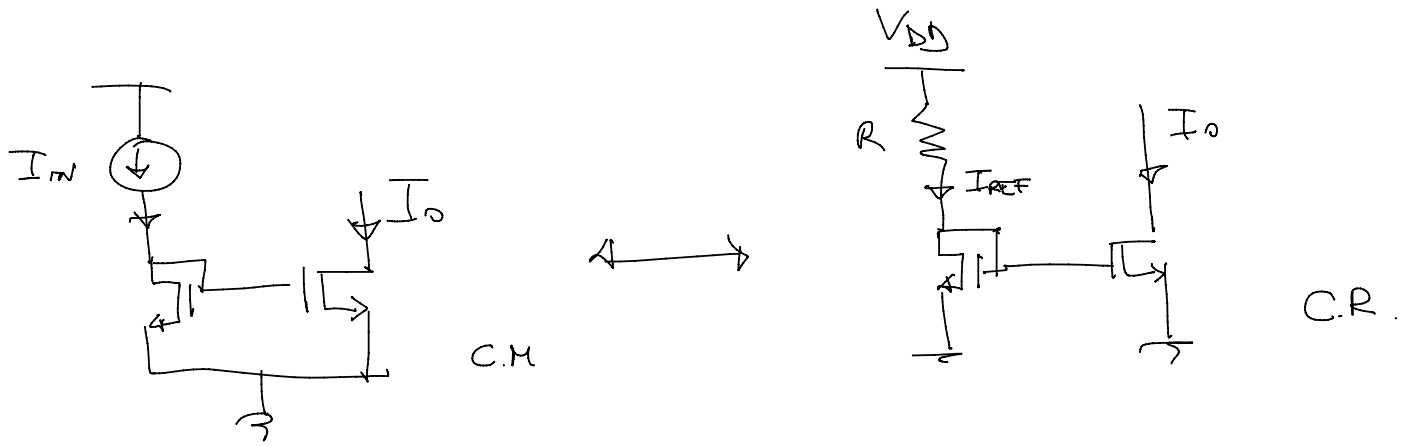
R_{oCM} is large !!

$$\underline{i_s = 0} \Rightarrow I_{IN} = \text{const}$$



$$I_O = \text{const}$$

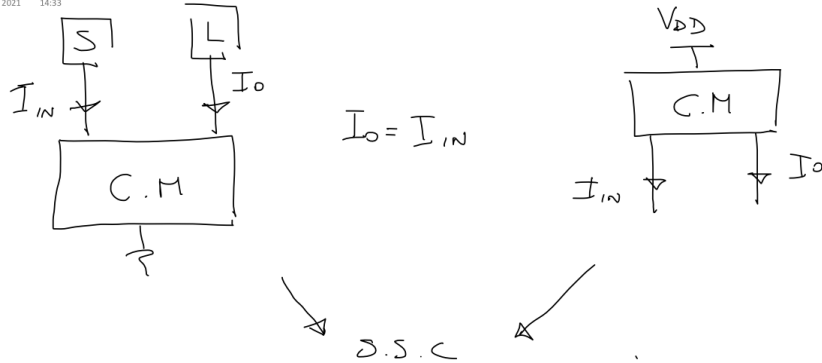
Current mirror $\rightarrow$ Current reference



తేపంతే తేల్చి పన్నుల

Brief Recap of Current Mirror Concepts (00:00 -

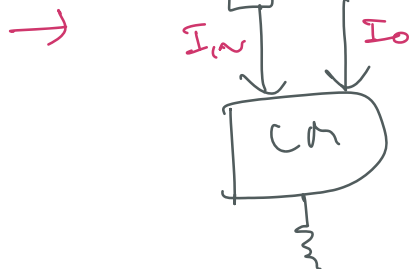
current mirrors and current references
lunedì 29 marzo 2021 14:33



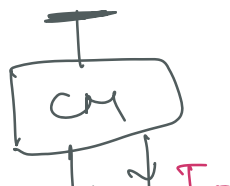
→ Generic current mirror can be represented by the Black Boxes shown as above. In which we have a certain i/p provided by a current - The role of the current mirror is to mimic the I/p current to the o/p Node.

i.e. $I_O = I_{IN}$

(o/p current should be as close as possible to the i/p current)



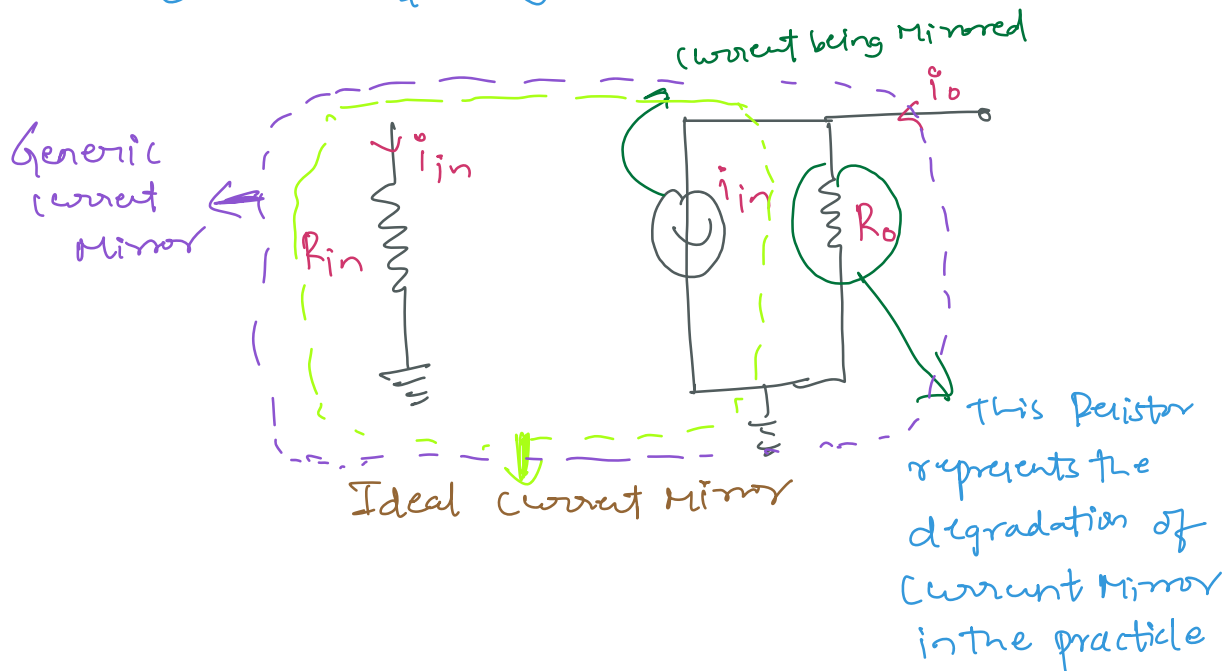
current mirror realized with PMOS Transistors



current mirror implemented using PMOS Transistors.

$$I_{in} \downarrow \uparrow +0$$

→ Both the above Current Mirror Configurations have the Same Small Signal Representation characterized by same Topology

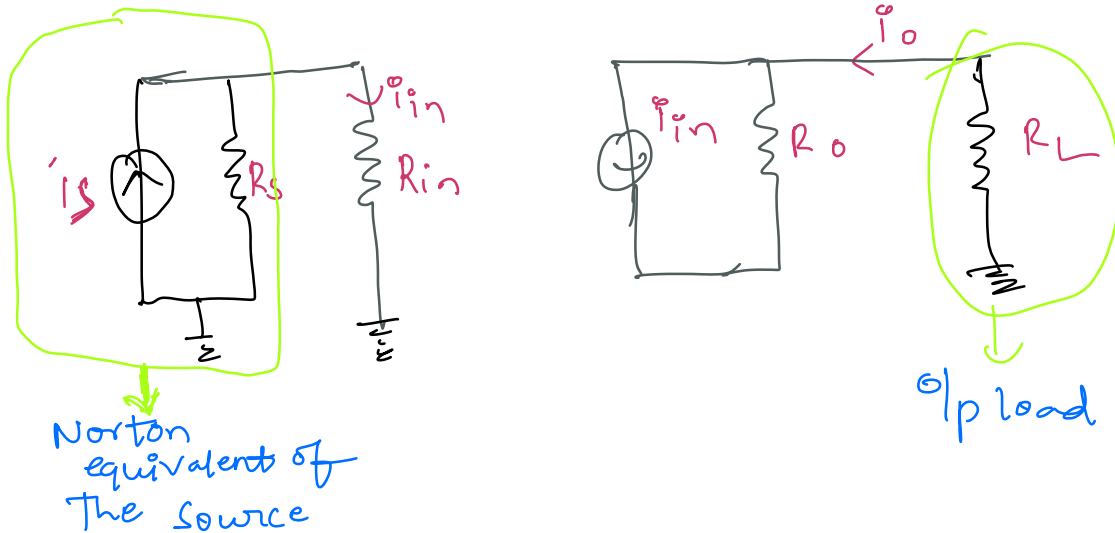


* Note:- When ever we look at case
SSM of a circuit we are only interested in the variations of the Quantity that we wish to control @ measure.

→ The above SSM is the "Generic representation" of the Current Mirror.

But the above topology is never alone it is always chained with processing stages with electrical quantities we are consider.

For example! current mirror i/p stage is
always connected to a current
source.



→ By taking into account all the Nonidealities ^{as much as possible}
we would like to transfer ^{as much as possible} the information
carried by this current I_S to the load R_L .

But this is True if

$$R_S \gg R_{in_{cm}}$$

$$R_S \gg R_{in_{cm}}$$

$$i_{in} = \frac{R_S}{R_S + R_{in_{cm}}} i_S$$

Small!

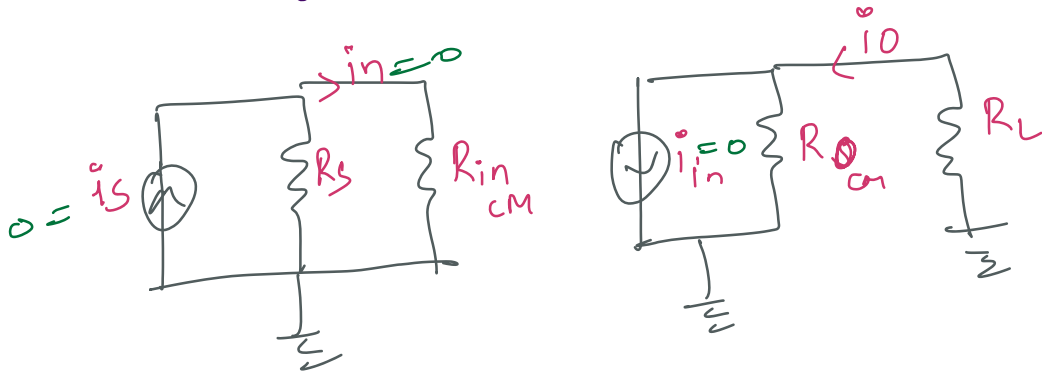
$$R_{O_{cm}} \gg R_L$$

$$R_{O_{cm}} \gg R_L$$

$$i_o = \frac{R_{O_{cm}}}{R_{O_{cm}} + R_L} i_{in}$$

\$R_{O_{cm}}\$ is large !!

→ I_S $I_S = I_{ref}$ $i_S = 0 \Rightarrow$ that constant

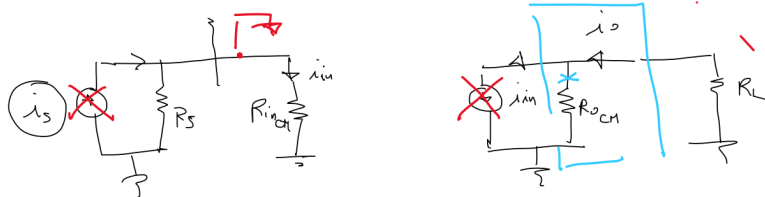


If $i_S = 0 \Rightarrow i_{in} = 0$, then there is no current to be mirrored.

→ If there is no current to be mirrored the only thing that matters is o/p Node.

∴ Current Mirror with $i_S = 0$ is consistent with o/p Resistance $R_{O_{cm}}$

$i_S = 0$ $\Rightarrow I_{IN} = \text{const}$

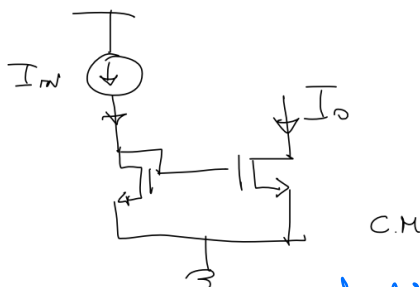


$I_O = \text{const}$ (If $i_S = 0$ then $i_{in} = 0 \Rightarrow i_O = 0$ that's variations in o/p is zero if variations in i/p are also zero)

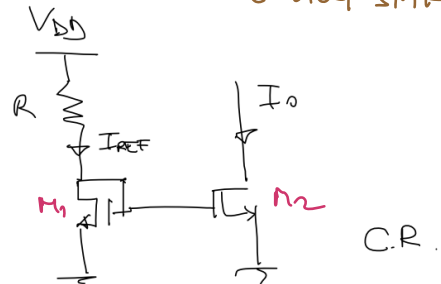
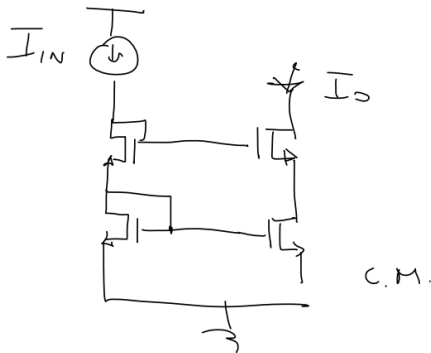
$\therefore$ o/p current is constant

*# When $i_s = 0$ rather than having Current Mirror we are having Current Reference (or) Const current Generator

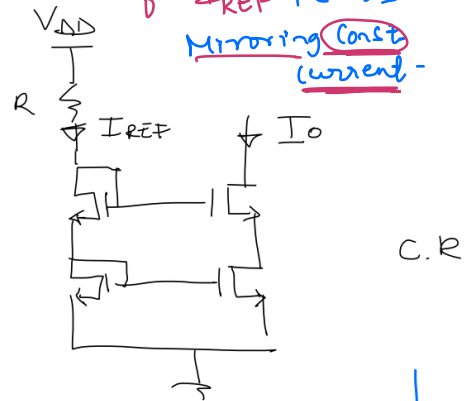
Current mirror $\rightarrow$ Current reference (or) Current Generator (or) Current Sink



Current mirror duplication
Generic $1/p$ current.



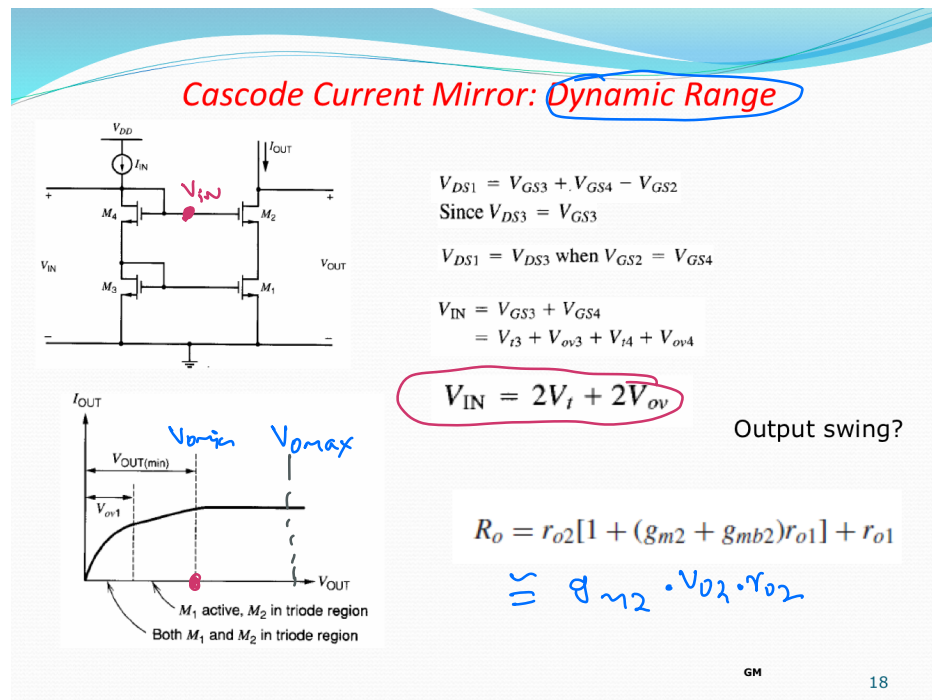
If both M_1, M_2 are same then I_O is the Duplication of I_{REF} i.e. it is Mirroring Const current.



$\rightarrow$ Do you remember why cascode current mirror might be but selection in some application?

It is useful when we want high output Resistance.

However, Cascode may have some problems too.

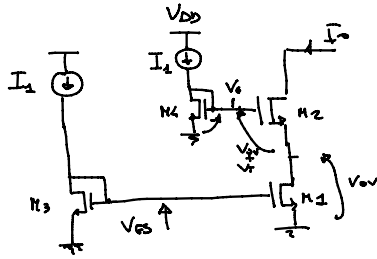


→ Dynamic Range is the range of output voltage for which I_o will remain constant.

and $V_{omin} = 2V_{ov} + V_t$ This quantity can be large.

Threshold could be V_t which we derived in the earlier class

→ V_{omax} is decided by the load attached to the op.



$R_O \rightarrow \text{😊}$

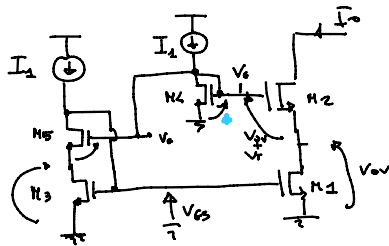
Swing ($V_{O,MIN} = 2V_{OV}$) $\rightarrow \text{😊}$

Power $\rightarrow \text{😞}$

Accuracy $\rightarrow \text{😞}$

$$I_D = f(V_{GS}, V_{DS}) \quad \left\{ \begin{array}{l} \text{Weak dep. (SAT)} \end{array} \right.$$

$$V_{DS3} = V_T + V_{OV} \quad ; \quad \underline{\underline{V_{DS1} = V_{OV}}}$$



$$V_{DS3} \neq V_{GS3}$$

$$V_G = 2V_{OV} + V_T$$

$$M5 \rightarrow \left. \begin{array}{l} \frac{W}{L}_5 = \frac{W}{L}_2 \\ I_5 = I_2 \end{array} \right\} V_{OV5} = V_{OV2} = V_{OV}$$

$$V_{GS5} = V_T + V_{OV}$$

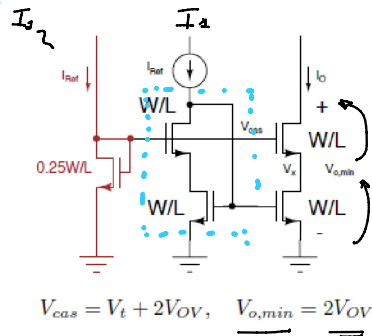
$$\underline{\underline{V_{DS3} = V_G - V_{GS5} = V_{OV}}}$$

Accuracy 😊

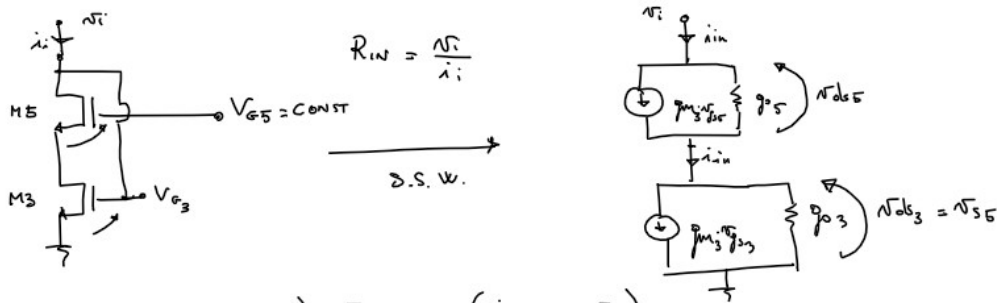
Cascode Bias Improved

The optimal V_{cas} is $V_T + 2V_{OV}$.

If we make a device 4 times smaller and keep the current constant, V_{OV} goes up by a factor of 2.



Sensitive to back-gate effect.



$$R_{in} = \frac{v_i}{i_i}$$

S.S.W.

$$v_{gs5} = -v_{s5} \quad ; \quad A) \quad v_{s5} = r_{o5} (i_i + g_{m5} v_{s5})$$

$$v_{gs3} = v_i \quad ; \quad B) \quad v_{s5} = r_{o3} (i_i - g_{m3} v_i)$$

$$C) \quad v_i = v_{s5} + v_{o5}$$

A) + B) + C)

$$\hookrightarrow v_i = r_{o5} (i_i + g_{m5} v_{s5}) + r_{o3} (i_i - g_{m3} v_i)$$

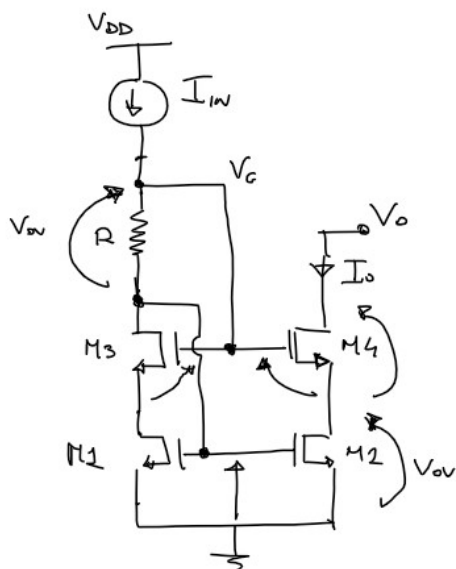
$$v_i = \underbrace{r_{o5}}_1 \cdot \underbrace{i_i}_3 + \underbrace{r_{o5} \cdot r_{o3} \cdot i_i \cdot g_{m5}}_4 - \underbrace{r_{o3} g_{m3} \cdot g_{m5} \cdot r_{o5} \cdot v_i}_2 + \underbrace{r_{o3} i_i}_6 - \underbrace{g_{m3} v_{o3} \cdot v_i}_5$$

$$1 \ll 2 \quad 5 \ll 2$$

$$3 \ll 4 \quad 6 \ll 4$$

$$\frac{v_i}{i_i} = \frac{\cancel{r_{o3}} \cdot \cancel{g_{m5}} \cdot \cancel{r_{o5}}}{\cancel{r_{o3}} \cdot \cancel{g_{m3}} \cdot \cancel{g_{m5}} \cdot \cancel{r_{o5}}} = R_{in} \approx \frac{1}{g_{m3}}$$

Self-biased cascode



$$V_G = 2V_{OV} + V_T$$

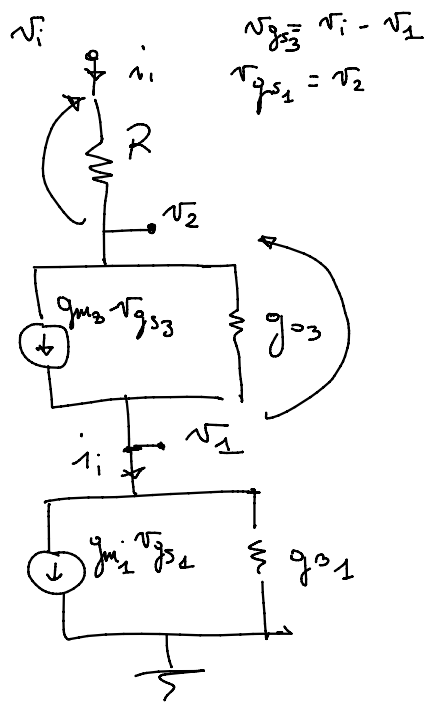
$$V_{D3} = V_{OV} + V_T = V_{G1,2}$$

$$V_R = V_G - V_{D3} = V_{OV} = R \cdot I_{in}$$

$$R_{out} = g_m \cdot r_o \quad \leftarrow \quad \frac{W}{L}_h = \frac{W}{L}_e$$

$$V_{O_{MIN}} = 2V_{OV}$$

$$\left. \begin{array}{l} V_{G1} = V_{G2} \\ V_{D1} = V_{D2} \end{array} \right\} I_O = I_{IN}$$

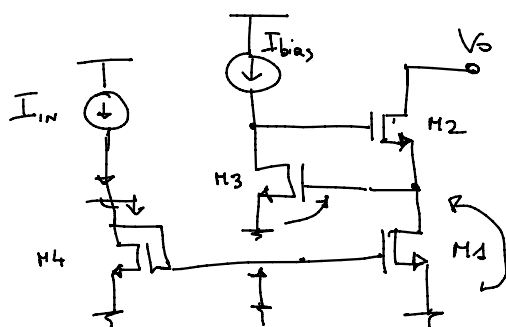


$$\begin{aligned}
 v_2 &= r_{o2} (i_1 - g_{m1} v_2) \\
 v_2 &= v_1 - R \cdot i_1 \\
 v_2 \cdot v_2 &= r_{o3} \cdot (i_1 - g_{m3} (v_1 - v_2))
 \end{aligned}
 \quad \Delta$$

$$\Delta \Rightarrow \frac{v_1}{i_1} = \frac{R + r_{o1} + r_{o3} + g_{m3} r_{o3} r_{o1} + g_{m1} \cdot r_{o2} \cdot g_{m3} \cdot r_{o3} \cdot R}{1 + g_{m1} r_{o1} + g_{m2} \cdot r_{o2} \cdot g_{m3} \cdot r_{o3} + g_{m1} \cdot r_{o1}}$$

$$\frac{v_1}{i_1} \approx \frac{1}{g_{m1}} + R = R_{in}$$

Regulated Cascode Mirror

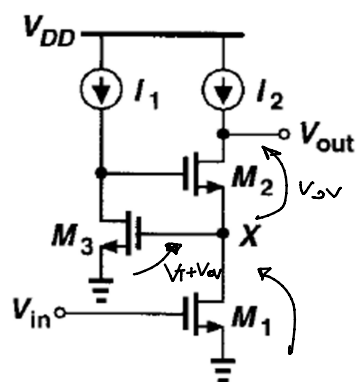


$$V_{GS4} = V_{GS1} \quad ; \quad V_{GS3} = V_{GS2} \Rightarrow V_{DS4} = V_{DS1}$$

Accuracy OK

$$R_{in} = \frac{1}{g_{m4}} \quad ; \quad R_{out} = g_{m1}^2 \cdot r_{o3}^3 \quad ; \quad V_{O_{MIN}} = \underline{\underline{2V_{ov} + V_T}}$$

Trans conductance amp



Current Mirror	Accuracy	R_{out}	R_{in}	Output Swing	Power
Simple	Poor	r_o	$\frac{1}{g_m}$	$V_{min} = \underline{\underline{V_{ov}}}$	
Cascode	Good	$r_o^2 \cdot g_m$	$\frac{2}{g_m}$	$2V_{ov} + V_T$	
High Swing	Good	$r_o^2 \cdot g_m$	$\frac{1}{g_m}$	$2V_{ov}$	(i)
Self Biased	Good	$r_o^2 \cdot g_m$	$\frac{1}{g_m} + R$	$2V_{ov}$	
Regulated Cascode	Good	$g_m^2 \cdot r_o^3$	$\frac{1}{g_m}$	$V_T + 2V_{ov}$	

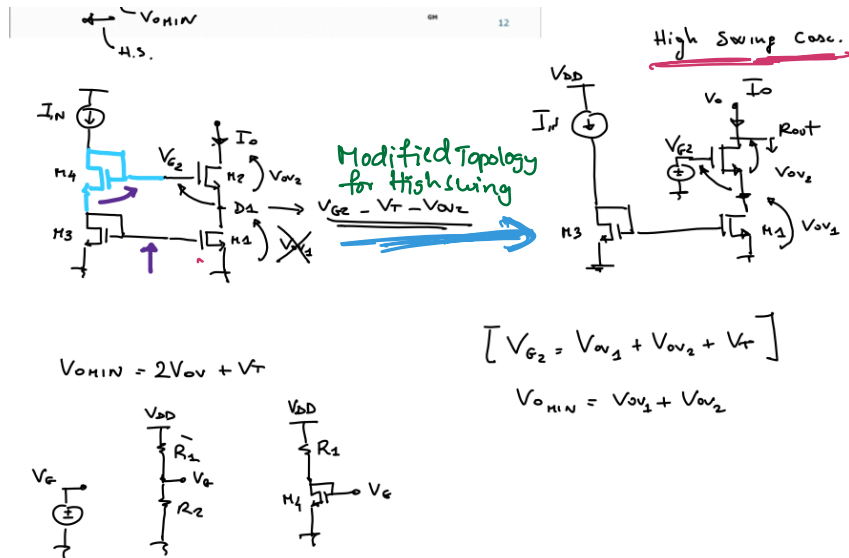
* Limitations of The Cascode Current Mirror *

→ Q What could be the better solution for the above problem?

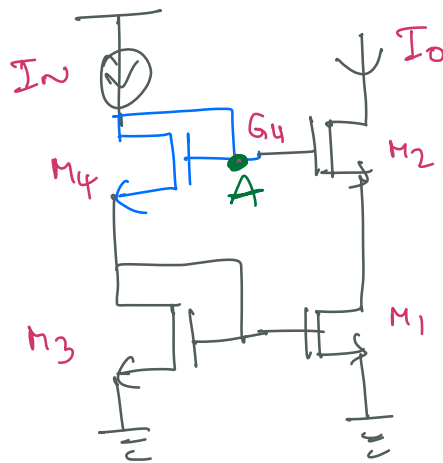
Q What are the true limitations of the Cascode stage Current Mirror?

Ans

* Advanced Current Mirrors *



→ Modified Topology for High Swing



- M_4 Transistor is used to feed Cascoding Transistor M_2 in the 2nd Branch
- M_3 is used to provide Gate voltage for the M_1 transistor.

The Problem of the Topology is mostly given by the fact that M_4 & M_3 are short circuited and

$$V_{G4} = V_{GS4} + V_{GS3}$$

$$= V_{OV} + V_T + V_{OV} + V_T$$

$$V_{G4} \Rightarrow 2V_{OV} + 2V_T$$

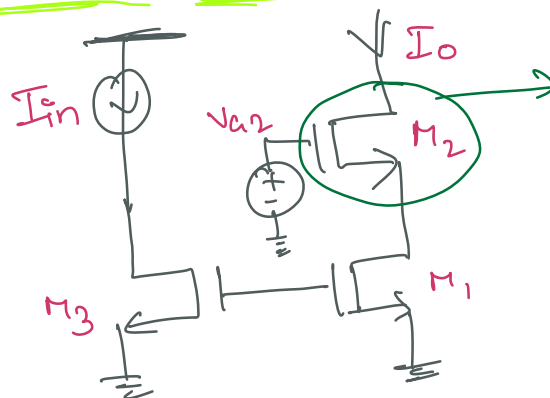
* This unnecessary high voltage for the node (A) is contributing to the low voltage margin for the current mirror

$$\therefore V_{Omin} = 2V_{OV} + V_T \quad \text{--- (A)}$$

∴ The idea to improve this schematic is to remove this Transistor M₄ which has no effect on the o/p cascode solution.

Because, cascode solution is given by Cascode effect of M₂ + M₁, not by the node voltage of (A).

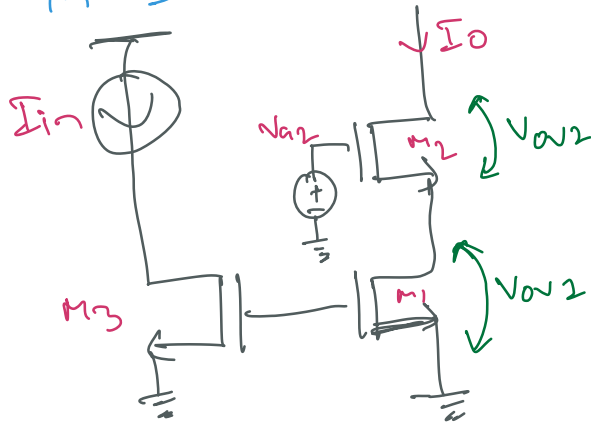
New Topology after removing M₄ High-Swing Cascode



This Common Gate is what gives us High Resistance which is what we want.

Q what would be Best value for V_{G2} ?

we want both M_1 & M_2 should be in Saturation



Now we are not constrained as in earlier case we can choose the value of V_{G2} .

$$\therefore V_{G2} = V_{O1} + V_{O2} + V_T$$

~~**~~

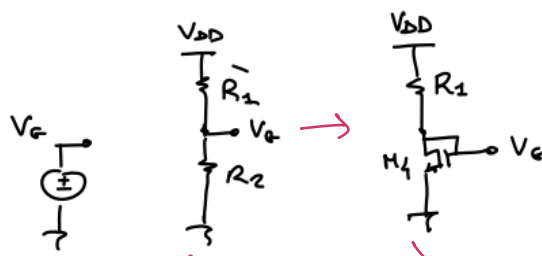
Now, minimum voltage required for the configuration is

$$V_{Omin} = V_{O1} + V_{O2} \quad \text{--- (b)}$$

which is less than the earlier cascode configuration. which was $V_{Omin} = V_{G1} + V_{O2} + V_T$

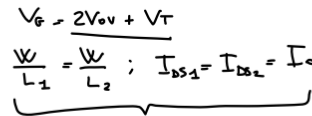
→ (a)

→ voltage Generator



In Von

✓



$$V_{ov3} = V_{ov}$$

$$V_G = \underline{2V_{ov} + V_T}$$

$$(I_{DS}, \frac{W}{L}, V_{OV})$$

2) $I_{DS4} = I_{DS3} = I_1$

$$I_{DS_x} = \frac{\beta_1}{2} \cdot \frac{W}{L_x} \cdot V_{OV_x}^2 \Rightarrow \cancel{\frac{\beta_1}{2}} \cdot \frac{W}{L_4} \cdot V_{OV_4}^2 = \cancel{\frac{\beta_1}{2}} \cdot \frac{W}{L_3} \cdot V_{OV_3}^2$$

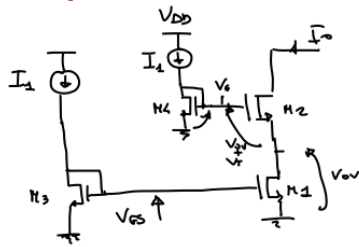
$$\frac{W}{L_3} = \frac{W}{L_1} = \frac{W}{L_2}$$

$$\frac{W}{L_4} \cdot 4 = \frac{W}{L_3}$$

Size Comparison of M_3 & M_4



High Swing Current Mirror



$R_o \rightarrow \text{😊} \checkmark$

Swing ($V_{o_{min}} = 2V_{ov}$) $\rightarrow \text{😊} \checkmark$

Power Consumption $\rightarrow \text{😞} \times$

Accuracy $\rightarrow \text{😞} \times$

$$I_D = f(V_{GS}, V_{DS}) \quad \left\{ \begin{array}{l} \text{Weak dep. (SAT)} \end{array} \right.$$

$$V_{DS} = V_T + V_{ov} \quad ; \quad \underline{V_{DS1}} = V_{ov}$$

$\rightarrow$ The overdrive of M_f is twice of all the other transistors.

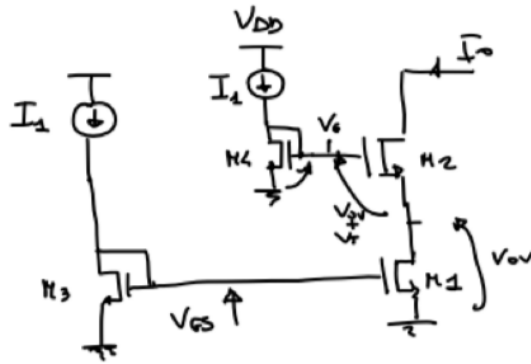
$\therefore$ In order to compensate for this effect we have size of M_f four times less than other transistors.

$\rightarrow$ The Fundamental Limit of the circuit is we have added an other Branch that means we have "More power consumption".

Accuracy of this solution is not okay for this circuit.

$$I_D = f(V_{GS}, V_{DS}) \quad \text{in saturation}$$

weak dependence



→ In this circuit to mirror input current I_{in} into I_O

$$I_{DS_{M3}} = I_{OS_{M1}} \\ (I_{in}) \quad (I_O)$$

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For both Transistors we have same $\frac{W}{L}$ & V_{GS}

But the only difference is

$$V_{DS3} = V_{OV} + V_T$$

$$\therefore V_{GS1} = V_{DS3} \\ (V_{OV} + V_T = V_{DS3})$$

~~**~~ while V_{DS1} of Transistor is equal to overdrive

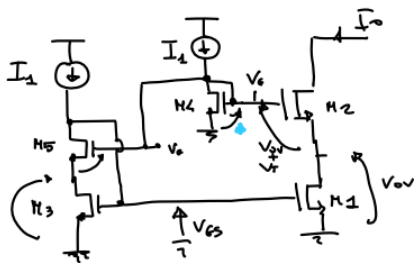
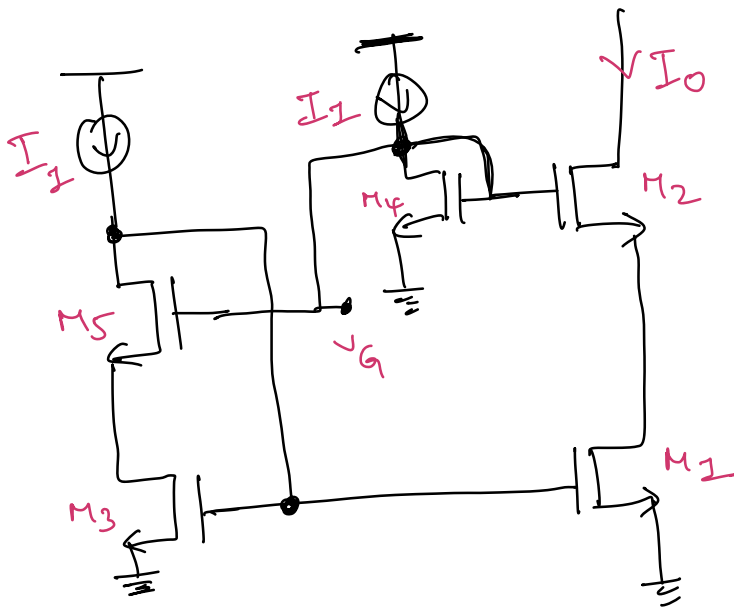
$$V_{DS2} = V_{OV}$$

This difference is not

Really okay even though Vps have a weak impact on I_{De} in saturation.

* we need a solution to minimize this discrepancy.

$\therefore$ we will make a change to the circuit topology.



Accuracy



$$V_{DS1} \neq V_{GS3} \quad (\text{First of all } V_{DS1} \neq V_{GS3})$$

$$V_G = 2V_{OV} + V_T$$

$$NS \rightarrow \left. \begin{aligned} \frac{W}{L_5} &= \frac{W}{L_3} \\ I_5 &= I_3 \end{aligned} \right\} V_{ov5} = V_{ov3} \neq V_{ov}$$

$$V_{GS5} = V_T + V_{OV}$$

$$V_{DS3} = V_G - V_{GS5} = V_{OV}$$

$$\therefore V_{DS3} = V_{DS1} = V_{OV}$$

Now the current running
is accurate"

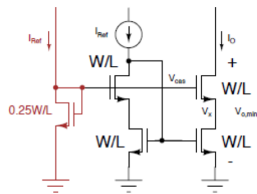
~~*~~



Cascode Bias Improved

The optimal V_{cas} is $V_t + 2V_{OV}$.

If we make a device 4 times smaller and keep the current constant, V_{ov} goes up by a factor of 2.



$$V_{cas} = V_t + 2V_{OV}, \quad V_{out,min} = 2V_{OV}$$

Sensitive to back-gate effect.

HEK

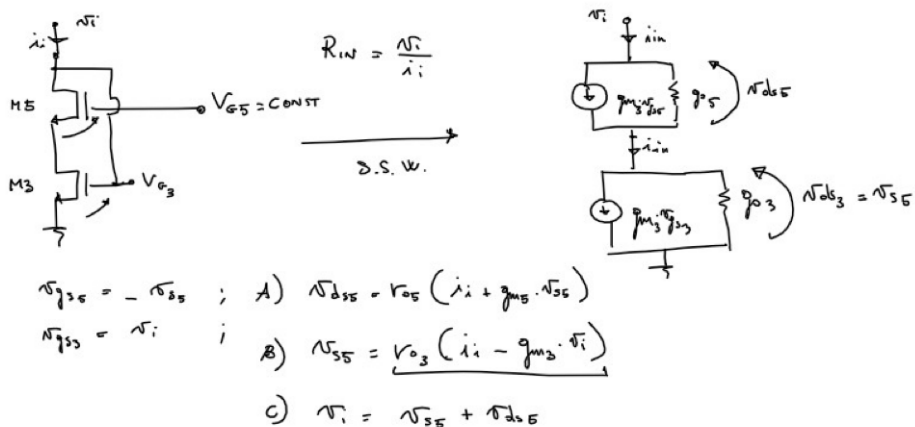
20

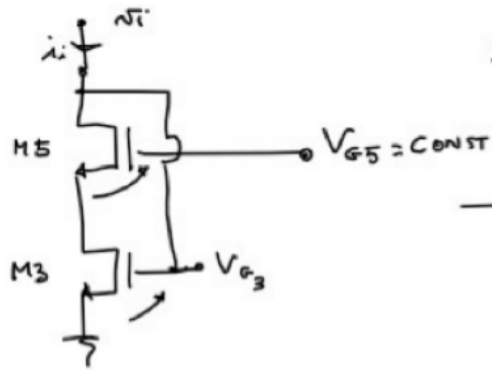
* For current mirrors we figured out that

i/p Impedance should be less.

o/p Impedance should be More.

→ let's study the i/p & o/p impedance of above circuit-



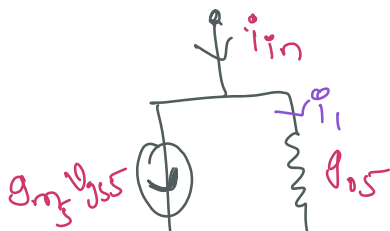


Here $V_{GS} = \text{constant}$
because V_{GS} is connected
to the Gate and Drain of
 M_4 and voltage
across is constant because
it is build as a voltage
divider with constant current.



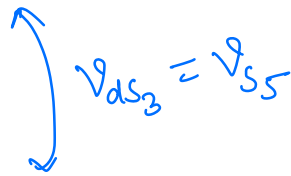
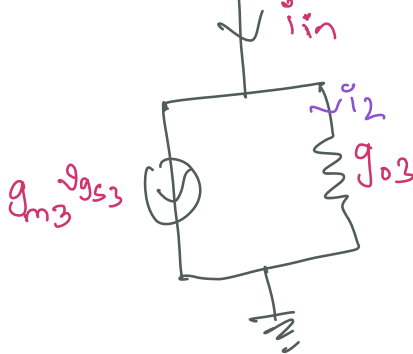
$$\therefore V_{GS5} = -V_{SS}$$

$$V_{GS3} = V_i$$



$$i_1 = \frac{V_{DS5}}{r_{O5}} = i_{in} - g_{m5} V_{GS5}$$

$$V_{DS5} = r_{O5} (i_{in} + g_{m5} V_{SS})$$



$$i_2 = i_{in} - g_{m3} V_{GS3}$$

$$\frac{V_{DS3}}{r_{O3}} = i_{in} - g_{m3} V_i$$

$$V_{SS} = r_{O3} (i_{in} - g_{m3} V_i)$$

$$V_{GS3} = V_i$$

$$\begin{aligned}
 * \quad v_{gs5} &= -v_{ds5} \quad ; \quad A) \quad v_{ds5} = r_{o5} (i_i + g_{m5} \cdot v_{gs5}) \\
 v_{gs5} &= v_i \quad ; \quad B) \quad v_{gs5} = \underbrace{r_{o3} (i_i - g_{m3} \cdot v_i)}_{\text{}} \\
 C) \quad v_i &= v_{gs5} + v_{ds5} \quad **
 \end{aligned}$$

calculation to determine Input Impedance

A) + B) + C)

$$\hookrightarrow v_i = r_{o5} (i_i + g_{m5} \cdot v_{gs5}) + r_{o3} (i_i - g_{m3} \cdot v_i)$$

$$\underbrace{v_i}_{1} = \underbrace{r_{o5} \cdot i_i}_{3} + \underbrace{r_{o5} \cdot r_{o3} \cdot i_i \cdot g_{m5}}_{4} - \underbrace{r_{o5} g_{m3} \cdot g_{m5} \cdot r_{o5} \cdot v_i}_{2} + \underbrace{r_{o3} i_i}_{6} - \underbrace{g_{m3} r_{o3} \cdot v_i}_{5}$$

$$1 \ll 2 \quad 5 \ll 2$$

$$3 \ll 4 \quad 6 \ll 4$$

$$\frac{v_i}{i_i} = \frac{\cancel{r_{o3}} \cdot \cancel{g_{m5}} \cdot \cancel{r_{o5}}}{\cancel{r_{o3}} \cdot \cancel{g_{m3}} \cdot \cancel{g_{m5}} \cdot \cancel{r_{o5}}} = R_{in} \approx \frac{1}{g_{m3}}$$

Input Impedance
is very low.

R_{out} is High

R_{in} is low for this circuit

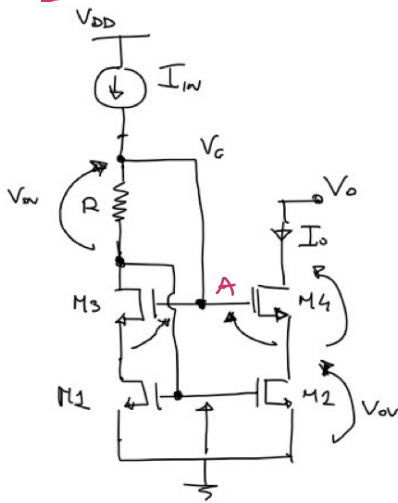
Topology

"only issue is power consumption ~"

~~***~~ In order to overcome this power consumption issue we have another circuit called

Self-Bias circuit ~~***~~

Self-biased cascode



$$V_G = 2V_{OV} + V_T$$

$$V_{D3} = V_{OV} + V_T = V_{G1,2}$$

$$V_R = V_G - V_{D3} = V_{OV} = R \cdot I_{IN}$$

$$R_{out} = g_m \cdot r_o^2 \rightarrow \frac{W}{L}_h = \frac{W}{L}_l$$

$$V_{O_{MIN}} = 2V_{OV}$$

$$\left. \begin{array}{l} V_{G1} = V_{G2} \\ V_{DS1} = V_{DS2} \end{array} \right\} I_O = I_{IN}$$

→ Self Bias means we don't need additional device to Bias the circuit.

→ First we want all transistors to be in saturation and we first assume $V_{DS2} = V_{OV}$

⇒ That the voltage at Node A is V_G

$V_G = 2V_{OV} + V_T$ to put M_4 in saturation.

M_3 should also be in saturation
But

$$\therefore V_{D3} = V_{G1,2} \\ = V_{OV} + V_T$$

→ $\therefore$ The voltage across Resistor is equal to

$$V_R = V_G - V_{D3} = 2V_{OV} + V_T - (V_{OV} + V_T)$$

$$V_R \Rightarrow V_{OV} = R \cdot I_{in}$$

→ The first Branch in this circuit is just used for Biasing the second Branch.

Then $R_{out} \approx g_m \cdot r_o^2$
↓
approximated $\sim$ r_p Resistance

→ $\therefore V_{Omin} = 2V_{OV}$ ← assuming $\left(\frac{W}{L}\right)_4 = \left(\frac{W}{L}\right)_2$

*

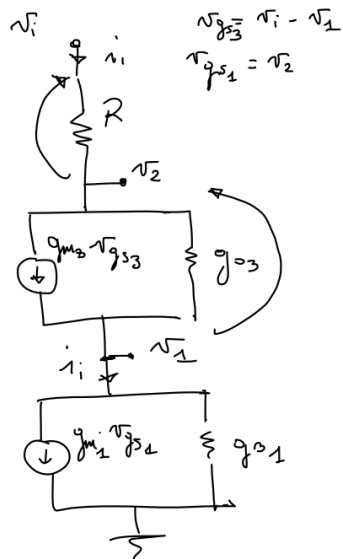
$$V_{GS1} = V_{GS2}$$

Then $I_0 = I_{in}$
Faithful Replication

$$V_{DS1} = V_{DS2}$$

Small signal Model of The Model

calculation of input Resistance



$$\begin{aligned}
 v_o &= r_{o2} (i_i - g_{m1} v_1) \\
 v_1 &= v_i - R \cdot i_i \\
 v_o \cdot v_1 &= r_{o2} \cdot (i_i - g_{m1} (v_i - v_1))
 \end{aligned}
 \quad \text{--- } \textcircled{\Delta}$$

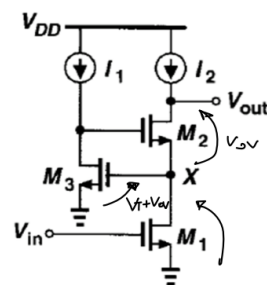
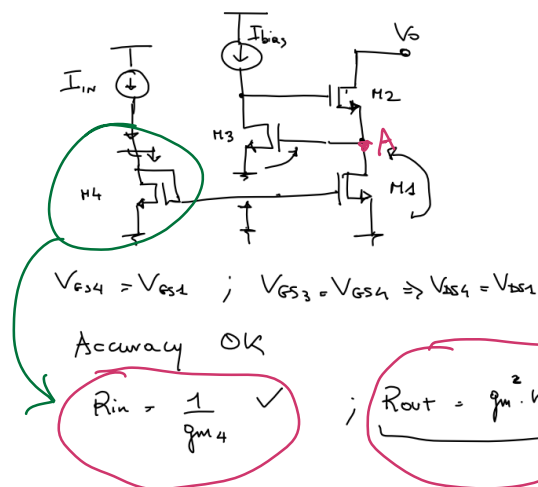
$$\textcircled{\Delta} \Rightarrow \frac{v_i}{i_i} = \frac{R + r_{o2} + r_{o3} + g_{m3} r_{o3} r_{o2} + g_{m1} \cdot r_{o2} \cdot g_{m3} r_{o3} \cdot R}{1 + g_{m1} r_{o3} + g_{m2} r_{o2} \cdot g_{m1} r_{o3} + g_{m1} r_{o2}}$$

$$\frac{v_i}{i_i} \approx \frac{1}{g_{m1}} + R = R_{iu}$$

* Regulated Cascode Mirror

Regulated Cascode Mirror

Transconductance amp



Very large
o/p Resistance
(Has already been
solved)

swing is
not OK

$$V_{GS4} = V_{GS1}$$

obvious from circuit

$$V_{GS4} = V_{DS4}$$

since Gate and Drain are shorted,
for M_4

$$V_{GS3} = V_{DS1} \text{ from Node A}$$

If M_3 can be sized as M_4

$$\Rightarrow V_{GS3} = V_{GS4} \text{ then}$$

$$V_{DS4} = V_{DS1}$$

Current Mirror	Accuracy	R_{out}	R_{in}	Output Swing
Simple	Poor	r_o	$\frac{1}{g_m}$	$V_{min} = \underline{V_{ov}}$
Cascode	Good	$r_o^2 \cdot g_m$	$\frac{2}{g_m}$	$2V_{ov} + V_T$
High Swing	Good	$r_o^2 \cdot g_m$	$\frac{1}{g_m}$	$2V_{ov}$
Self Biased	Good	$r_o^2 \cdot g_m$	$\frac{1}{g_m} + R$	$2V_{ov}$
Regulated Cascode	Good	$g_m^2 \cdot r_o^3$	$\frac{1}{g_m}$	$V_T + 2V_{ov}$

Power

①

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